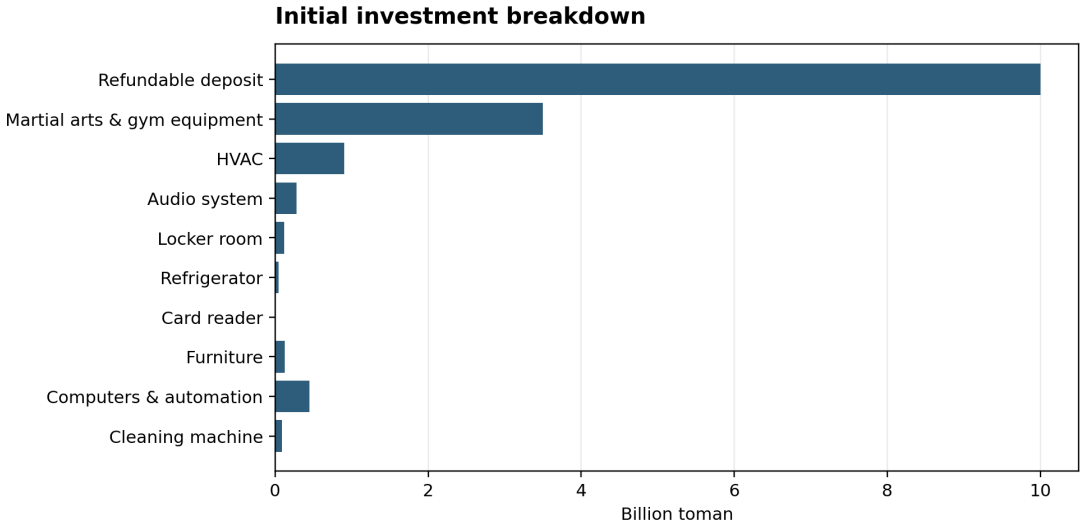
INDUSTRIAL ENGINEERING | ENGINEERING ECONOMICS

Financial Feasibility Analysis of a Multi-Purpose Sports Club

A model-based assessment of investment, operating cash flows, financing, and economic risk



Research summary based exclusively on the supplied project report and Excel model

Initial investment: 15.523 billion toman | Base-case NPV: 3.925 billion toman
Reported real IRR: 45.1% | Payback period: 4.96 years

Industrial Engineering Department
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Prepared as a structured English companion to the original Persian course project

Executive Summary

This report summarizes the financial feasibility study of a proposed multi-purpose sports club. The original project combines a detailed Excel model with a Persian written report covering initial investment, operating expenses, class-based revenue, beverage sales, depreciation, tax treatment, a five-year bank loan, ten-year cash-flow projections, and sensitivity analysis.

The model estimates an initial investment of **15.523 billion toman**, including a **10 billion toman refundable property deposit** and equipment/facility assets. The main operating revenue source is assumed to be sports classes and memberships; beverage sales are a smaller supplementary source. First-year class revenue is estimated at **19.7142 billion toman**, while the modelled first-year operating-cost total excluding beverage purchases is **12.209765 billion toman**.

Under the base-case assumptions, the report states a net present value (NPV) of **3.925 billion toman**, a reported inflation-adjusted or "real" internal rate of return (IRR) of approximately **45%**, and a payback period of **4.96 years**. The project uses a minimum attractive rate of return (MARR) of **37%**. These results indicate acceptability under the model's own assumptions, not a guarantee of actual commercial performance.

The sensitivity analysis is central to the study. The report shows that the NPV approaches zero when revenue falls to 60% of the base case, while a 20% cost increase makes NPV negative. In the two-factor analysis, the reported real IRR ranges from about 21% under an adverse combination of lower revenue and higher costs to about 94% under a favourable combination.

Interpretation boundary. This is a model-based academic feasibility study. It does not independently validate future demand, actual occupancy, price growth, or market conditions. Some cached Excel sensitivity cells do not reconcile with the corresponding sensitivity tables and narrative in the PDF; the sensitivity discussion in this summary follows the PDF's stated results, and those workbook tables should be recalculated and verified before external reliance.

1 Project Scope and Decision Question

The central decision question is whether the proposed sports club can generate cash flows sufficient to justify its initial investment under the project's assumed operating and economic conditions. The written report frames feasibility as a joint assessment of investment costs, operating expenses, revenue streams, financing, taxes, inflation, and risk rather than as a simple comparison of sales and expenses.

The planned facility is multi-purpose. The model includes a gym area and several class-based activities, including self-defense, taekwondo, functional boxing, kickboxing, jiu-jitsu, TRX, Pilates, and MMA. It also includes sales of energy drinks and bottled water. The asset plan includes gym and martial-arts equipment, HVAC, audio equipment, a locker room, reception-related equipment, computers/automation, and cleaning equipment.

The study horizon is ten operating years, preceded by a year-zero investment. It estimates annual nominal cash flows and evaluates them using NPV, IRR, payback, and sensitivity analysis. A bank loan is included as a financing component, with scheduled payments during the first five operating years.

The analysis therefore has four linked layers:

1. **Operational model:** class capacity, assumed participation, fees, and ancillary beverage sales.
2. **Cost model:** initial assets, rent, utilities, staffing, maintenance, cleaning, advertising/insurance, and beverage purchases.
3. **Financial model:** depreciation, beverage tax, debt service, inflation, deposit recovery, and terminal salvage assumptions.
4. **Decision model:** discounted cash flow, NPV, inflation-adjusted IRR, payback, and sensitivity scenarios.

The report's value lies in connecting these layers in a single spreadsheet model. Its conclusions must be interpreted in relation to the assumptions that drive each layer.

2 Analytical Workflow and Model Architecture

The workbook is organized into two worksheets: a main model and a loan-amortization schedule. The main worksheet contains assumptions, asset investment, operating costs, beverage revenue, class revenue, annual loan payments, ten-year cash flows, economic indicators, and sensitivity tables. The loan worksheet calculates monthly installments and separates interest from principal repayment.

The model follows this sequence:

Inputs and assumptions
↓
Initial investment + annual operating costs + revenue streams
↓
Inflation/escalation + beverage tax + loan repayment
↓
Annual nominal after-tax cash flows (years 0–10)
↓
NPV, real IRR, payback, and sensitivity scenarios

This architecture is appropriate for an engineering-economics case study because it makes the relationship between operational assumptions and financial outcomes explicit. It also creates a practical basis for scenario testing: revenue and cost multipliers can be changed while other assumptions are held constant.

The spreadsheet includes formula-based calculations, including straight-line depreciation, the monthly payment function for the loan, NPV, and IRR. However, model outputs depend on correct cell references and recalculation of Excel's what-if data tables. A spreadsheet that contains formulas is not automatically validated; consistency checks between the workbook and written report remain necessary.

3 Core Economic Assumptions

The main worksheet records the following headline assumptions. These are treated as model inputs rather than independently verified economic forecasts.

Table 1: Key assumptions recorded in the Excel model

Parameter	Model value
Apparent annual rate / discount factor used in NPV	106%
Minimum attractive rate of return (MARR)	37%
Average annual inflation assumption for goods	50.6%
Tax rate applied to beverage profit in the workbook	25%
Bank loan principal	15.0 billion toman
Annual loan interest rate	23%
Loan term	60 monthly payments
Evaluation horizon	10 operating years

The model does not apply a single escalation rate to every cost and revenue category. For example, it assigns 25% annual escalation to utility bills, 30% to personnel costs, 50.6% to maintenance and cleaning supplies, 35% to advertising and insurance, 40% to rent, and 40% to class tuition. Beverage prices and beverage-purchase costs are escalated using 50.6%.

These category-specific rates are intended to represent differing cost and revenue behaviour. At the same time, several of the assumed annual increases are high and compound over ten years. Consequently, long-term nominal cash flows are particularly sensitive to the escalation assumptions. Any interpretation of the ten-year results should make this dependence explicit.

4 Initial Investment and Asset Structure

The model estimates the year-zero investment at **15.523 billion toman**. The largest component is the 10 billion toman refundable property deposit. The remaining amount funds equipment and facility assets. The report's fixed-asset table uses straight-line depreciation for depreciable items, with asset lives ranging from three to ten years; the deposit is not depreciated.

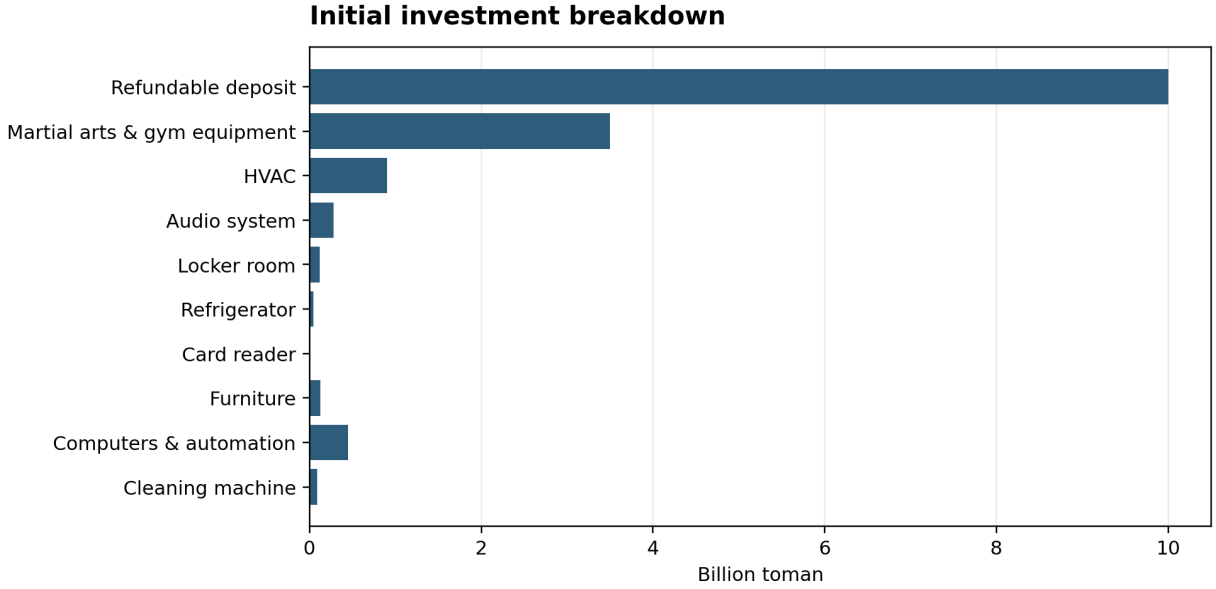


Figure 1: Initial investment components recorded in the Excel model; values are in billion toman.

Table 2: Initial investment components

Component	Toman
Refundable property deposit	10,000,000,000
Martial arts and gym equipment	3,500,000,000
HVAC system	900,000,000
Audio system	283,000,000
Locker room	120,000,000
Computers and automation	450,000,000
Furniture, refrigerator, card reader, cleaning machine	270,000,000
Total initial investment	**15,523,000,000**

The deposit accounts for roughly 64.4% of the initial investment. It is a major liquidity commitment, but the model assumes its return in year 10. The non-deposit investment is concentrated in gym and martial-arts equipment. This structure matters for risk: the deposit depends on contractual recovery, while equipment costs are exposed to purchasing, maintenance, and replacement risks.

5 Operating Cost Structure

The first-year operating-cost total recorded by the workbook is **12.209765 billion toman**, excluding beverage purchases in the cash-flow structure. The dominant item is annual rent, estimated at 12 billion toman. The other categories are much smaller in the model: utilities, personnel, maintenance, cleaning supplies, advertising/insurance, and beverage inventory.

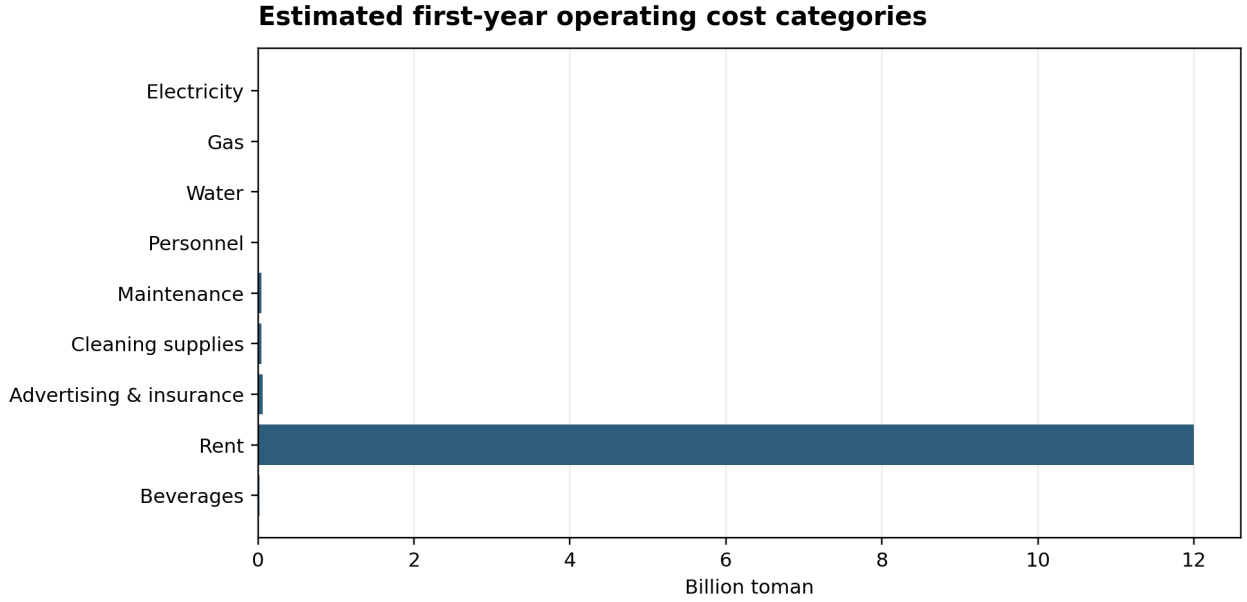


Figure 2: Estimated first-year operating cost categories from the workbook. Rent dominates the modelled operating-cost total.

The high concentration in rent means that lease terms are a central financial driver. Rent escalation is assumed at 40% per year. Under compounding, this assumption increases the rent burden quickly and materially affects long-term cash flow. It also means that the business case depends on the lease terms being realistic and stable enough to support the projected service revenue.

Maintenance and cleaning supplies are assigned annual escalation of 50.6%, while personnel costs are escalated at 30%. Utility expenses are escalated at 25%. These different rates influence the projected cost trajectory; the report's approach is more differentiated than applying one general inflation rate to all categories.

A practical managerial implication is that cost control should not be limited to small discretionary expenses. Lease negotiations, utilization of the facility, preventive maintenance, energy management, and staffing arrangements are likely to have much greater impact than minor savings in individual consumables.

6 Utility and Facility-Cost Estimation

The original Persian report documents separate estimation approaches for electricity, gas, and water. Electricity use is built from equipment power and assumed daily usage. The report estimates average daily electricity consumption at about 40.6 kWh and monthly consumption at approximately 1,238.3 kWh. It then applies a weighted energy tariff, demand charges, a fixed subscription charge, seasonal adjustments, and an additional tax/levy allowance. The resulting first-year electricity estimate is about 9.355 million toman.

The gas estimate separates hot-water demand for showers from winter space heating. Under the report's assumptions, the annual first-year gas cost including the stated charges is about 16.1 million toman. Water is modelled separately, with the spreadsheet using a first-year estimate of approximately 3.954 million toman. The report presents the utility calculations as estimates based on the tariffs and assumptions documented in the original project.

These amounts are small compared with rent and equipment investment, but the methodology is important. Utility estimates are built from usage assumptions rather than a verified operating history for this specific facility. The actual bill would depend on occupancy, equipment efficiency, weather, tariff changes, operating hours, and the mix of services offered.

The estimates should therefore be treated as transparent engineering approximations. A future version could improve accuracy by tracking actual meter readings from a comparable club and testing low, base, and high usage scenarios. The original report's separate treatment of electricity, gas, and water provides a clear starting structure for that refinement.

7 Revenue Model: Sports Classes

The spreadsheet treats sports classes as the principal revenue source. It includes ten activities: self-defense, taekwondo, functional boxing, gym access without a trainer, gym access with a trainer, kickboxing, jiu-jitsu, TRX, Pilates, and MMA. The estimated first-year class revenue totals **19.7142 billion toman**. Class fees and participation assumptions are set individually, allowing revenue to be built up from different service lines.

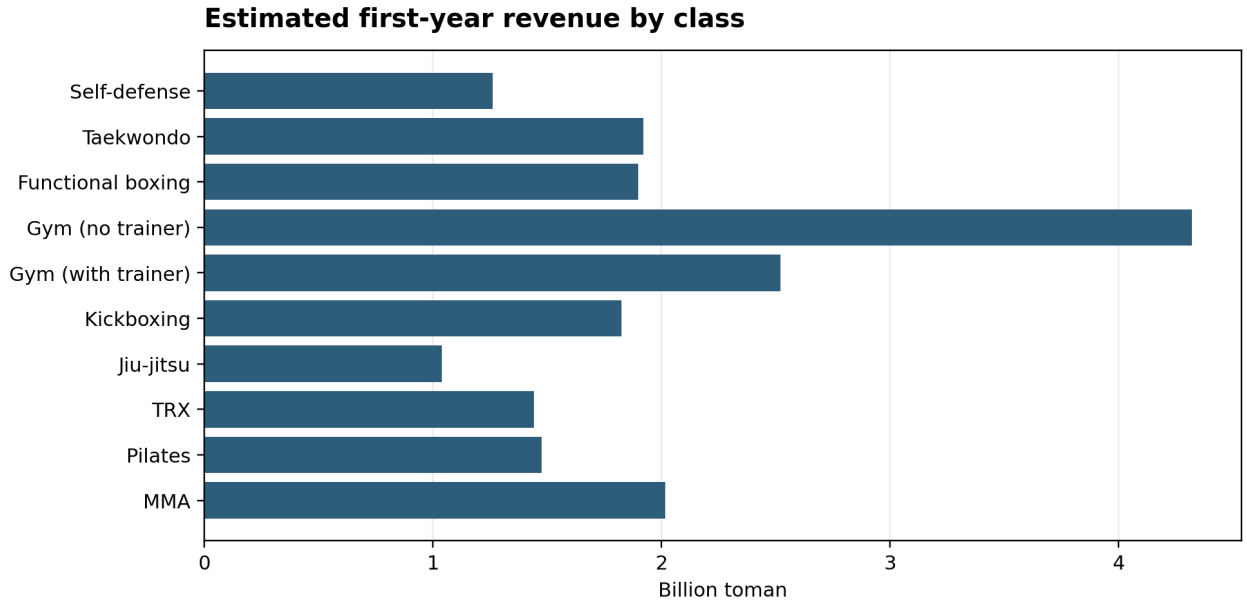


Figure 3: Estimated first-year revenue by class, based on the workbook's participation and fee assumptions.

The largest class-revenue line is gym access without a trainer at 4.32 billion toman. Gym access with a trainer contributes 2.52 billion toman, while MMA contributes 2.016 billion toman and taekwondo 1.92 billion toman. The model therefore does not rely on one single class; it assumes a mix of fitness and martial-arts activities.

The model uses a 40% annual increase in class tuition. This is a major assumption, and its compounding effect is one reason the nominal cash flows rise sharply in later years. The model's sensitivity analysis changes total revenue to assess how much the investment case depends on achieving the assumed revenue path.

Operationally, the revenue plan depends on the number of paying participants, class utilization, retention, trainer availability, scheduling, and the willingness of customers to pay the assumed monthly fees. The spreadsheet estimates these values; it does not demonstrate that these attendance and pricing levels have already been achieved in a real operating club.

8 Supplementary Revenue: Beverage Sales

The workbook includes sales of energy drinks and bottled water as supplementary revenue streams. It assumes a first-year unit price of 140,000 toman for energy drinks and 20,000 toman for bottled water. Under the workbook's stated quantities, annual sales revenue is estimated at 252 million toman for energy drinks and 72 million toman for bottled water, or 324 million toman in total.

The model also includes beverage purchase costs of 20 million toman in the first year. A 25% tax rate is applied to the difference between beverage sales and purchase costs. At the base-case values, the model calculates 304 million toman of beverage revenue net of purchase cost, 76 million toman of tax, and 228 million toman of after-tax beverage profit.

Table 3: First-year beverage model

Item	Toman
Energy-drink sales	252,000,000
Bottled-water sales	72,000,000
Total beverage sales	324,000,000
Beverage purchase cost	20,000,000
Taxable margin before tax	304,000,000
Tax at 25%	76,000,000
After-tax beverage contribution	**228,000,000**

Compared with class revenue, beverage sales are a secondary source of cash generation. The project appropriately separates beverage sales from class fees because beverage purchases and the stated tax treatment affect the net contribution. However, the calculation depends on the assumed sales volume, purchase price, and tax treatment remaining representative of the operating environment.

9 Financing Structure and Loan Amortization

The model includes a bank loan of **15 billion toman** at an annual nominal interest rate of **23%**, repaid over **60 months**. The Excel workbook uses the payment function to calculate a monthly installment of approximately **422.86 million toman**. The resulting annual debt service is about **5.074 billion toman** during each of the first five operating years.

Table 4: Annual loan schedule reported in the workbook

Year	Total payments	Interest	Principal repaid
1	5.074 bn	3.267 bn	1.807 bn
2	5.074 bn	2.805 bn	2.269 bn
3	5.074 bn	2.224 bn	2.850 bn
4	5.074 bn	1.495 bn	3.579 bn
5	5.074 bn	0.579 bn	4.495 bn

The payment remains approximately constant, while the interest component declines as the outstanding principal is repaid. This is consistent with an amortizing loan structure. Debt service has a major impact on cash availability in the first five years, particularly during the early stage of customer acquisition.

A key modelling distinction is that debt financing changes the timing and distribution of cash flows; it does not remove the underlying operating risk. If revenue underperforms or costs exceed assumptions during the loan period, the club still has to meet scheduled payments. The original report therefore emphasizes cash planning and the need for sufficient liquidity during the initial operating years.

10 Depreciation, Tax, and Terminal Items

The workbook applies straight-line depreciation to depreciable assets and assigns different useful lives to the asset categories. Examples include ten years for gym equipment and HVAC, six years for the audio system, five years for the card reader, three years for computers and automation, and four years for the cleaning machine. The property deposit is not depreciated.

The spreadsheet's first-year depreciation amounts are calculated from asset cost, assumed salvage value, and useful life. Depreciation is a non-cash accounting expense, so its role in a cash-flow model is distinct from a cash purchase. The original report includes depreciation and tax discussions, while the cash-flow worksheet separately identifies beverage profit after the modelled beverage tax.

The terminal-year model includes two important items: a zero salvage amount in the displayed year-10 cash-flow row and a 10 billion toman return of the property deposit. Thus, the cash-flow model explicitly recognizes the deposit recovery at the end of the horizon, but it does not include a positive salvage value for the listed depreciable assets in that row.

The return of the deposit is an assumption rather than a guaranteed outcome. Its realization would depend on the lease contract, the condition of the property, and any contractual deductions or disputes. For decision-making, the deposit should be treated as a recoverable asset subject to contractual risk, not as an ordinary operating expense.

11 Cash-Flow Construction

The main worksheet constructs annual cash flow from year zero through year ten. Year zero contains the negative initial investment of 15.523 billion toman. For each operating year, the model combines class revenue, operating expenses excluding beverage purchases, after-tax beverage contribution, and annual loan payments. The final year also includes recovery of the property deposit.

In simplified form, the operating-year structure used in the spreadsheet can be represented as:

$$CF_t = R_t^{class} - C_t^{operating} + \Pi_t^{beverage,after-tax} - DebtService_t,$$

with the year-zero investment recorded as a negative cash flow and terminal deposit recovery added in year ten. This expression is a high-level description of the workbook structure, not a replacement for every cell-level formula.

The cash-flow series is nominal: revenues and cost categories are escalated using the assumptions assigned to each category. This is important because the annual values in later years are not expressed in constant purchasing-power terms. They reflect the model's compounded price and cost growth.

The spreadsheet uses NPV with the apparent annual rate entered as 106%. Separately, it computes a nominal IRR from the cash-flow series and applies the workbook's inflation-adjustment formula to report a real IRR of approximately 45.1%. These calculations are described as implemented in the source model; their interpretation depends on the consistency of the chosen discount-rate and inflation conventions.

12 Ten-Year Cash-Flow Projection

The workbook's projected after-tax cash flows increase substantially over the ten-year horizon. The year-one value is approximately 2.658 billion toman. The model then reports 23.433 billion in year two, 34.863 billion in year three, 50.877 billion in year four, and 73.315 billion in year five. The projected series rises further after the five-year loan payment period ends.

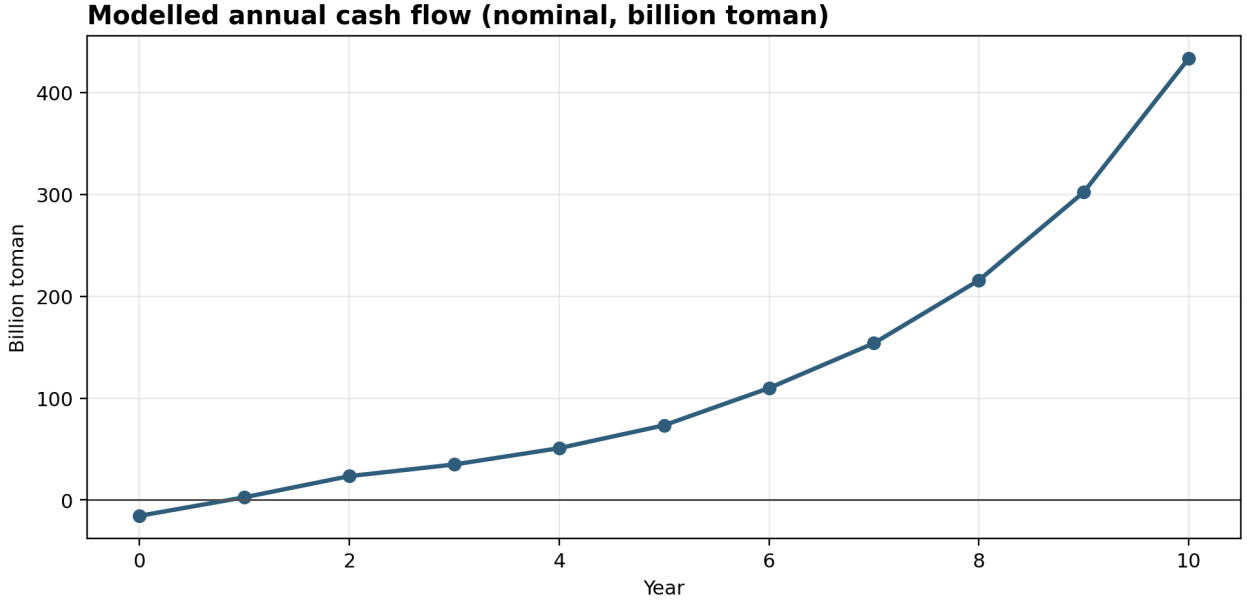


Figure 4: Nominal after-tax cash-flow series reproduced from the Excel model, including the year-zero investment.

Table 5: Modelled cash flows (billion toman)

Year	0	1	2	3	4
Cash flow	-15.523	2.658	23.433	34.863	50.877
Year	5	6	7	8	9–10
Cash flow	73.315	109.826	153.875	215.596	302.087 / 433.290

The trajectory should not be interpreted as a verified prediction. It is primarily the result of the assumed 40% annual tuition escalation, category-specific cost growth, and the model's revenue and utilization assumptions. The step-up after year five also reflects the end of scheduled loan payments. A useful next step would be to test a more conservative range of customer growth and price escalation assumptions and compare nominal cash flow with constant-price cash flow.

13 Net Present Value (NPV)

NPV measures the present value of future cash flows after deducting the initial investment. In the source workbook, the NPV formula discounts years one through ten at the apparent annual rate of 106% and adds the year-zero cash flow. Under that implementation, the reported base-case NPV is **3.925 billion toman**.

Base-case NPV
3.925 billion toman

A positive NPV indicates that the discounted cash flows exceed the initial investment under the selected discounting convention. It is not a statement that the club will necessarily produce that amount of cash in practice. The result is conditional on the projected revenue, cost escalation, loan schedule, terminal deposit recovery, and discount-rate assumption.

The report's revenue sensitivity analysis shows that NPV falls to approximately 50.2 million toman when total revenue is reduced to 60% of the base case. At a revenue multiplier of 1.40, the reported NPV rises to approximately 7.801 billion toman. The cost sensitivity analysis shows a sharper downside: a 10% increase in costs reduces NPV to approximately 560.6 million toman, while a 20% increase produces an NPV of about negative 2.804 billion toman.

The asymmetry matters. In the model, cost overruns can eliminate economic acceptability more quickly than comparable revenue changes. NPV is therefore useful not only as a headline result but also as a way to identify which assumptions need the strongest management controls.

14 Internal Rate of Return and Payback

The workbook calculates a nominal IRR of approximately 119.1% from the nominal cash-flow series. It then applies the workbook's stated inflation-adjustment formula, using the average inflation input of about 50.6%, to obtain a reported real IRR of approximately **45.1%**. The report compares this real IRR with the MARR of 37%.

Reported real IRR
45.1%

MARR
37%

On the report's stated basis, the real IRR exceeds the minimum acceptable return by approximately 8.1 percentage points. This supports a positive base-case decision under the project's chosen assumptions. The comparison should remain consistent: nominal IRR should not be compared directly with a real MARR without aligning the inflation basis.

The payback calculation uses the discounted cash-flow table and reports a period of **4.96 years**. This indicates recovery of the initial investment around the end of the fifth year under the report's calculation. Payback is easy to communicate but is less comprehensive than NPV because it does not fully summarize value created after the payback point. NPV and IRR should therefore be considered together, with payback used as a supplementary liquidity and recovery indicator.

The results are sensitive to both operating performance and the assumptions used to translate nominal cash flows into present values. The report's own risk section notes that the ability to attract and retain members, control operating costs, manage debt, and adapt to inflation will determine whether the projected return can be realized.

15 Sensitivity Analysis: Revenue

The report varies total revenue from 60% to 140% of the base case while holding the other model parameters constant. This is intended to isolate the effect of revenue performance on NPV. The source report's table is reproduced below through a chart.

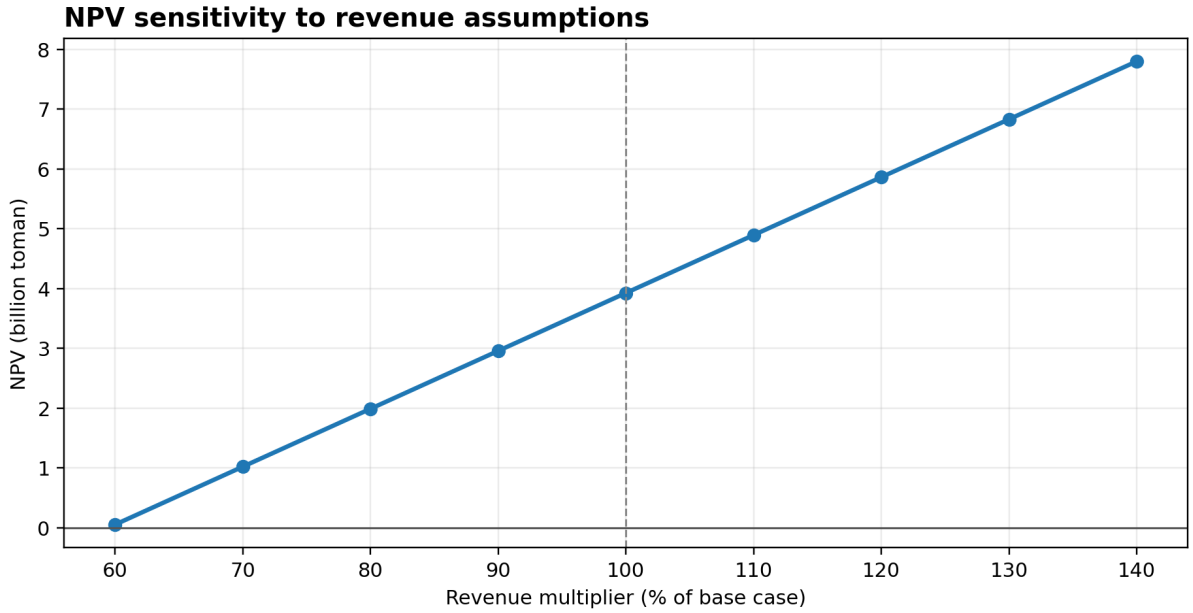


Figure 5: Reported NPV response to revenue multipliers. NPV values are in billion toman.

Table 6: Revenue sensitivity reported in the PDF

Revenue multiplier	NPV (billion toman)
0.60	0.050
0.70	1.019
0.80	1.988
0.90	2.957
1.00	3.925
1.10	4.894
1.20	5.863
1.30	6.832
1.40	7.801

The base case is economically positive, but the 60% revenue scenario leaves only a small positive NPV buffer. This suggests that a large decline in attendance, class participation, or realized fees could bring the project close to the break-even boundary. Revenue management priorities identified in the source report include member acquisition and retention, service quality, class variety, marketing, and better use of facility capacity.

16 Sensitivity Analysis: Costs

The report separately varies the cost multiplier from 60% to 140% of the base case, keeping the other assumptions unchanged. The resulting NPV response is more severe on the upside-cost side: a 10% increase in costs leaves a small positive NPV, while a 20% increase makes NPV negative.

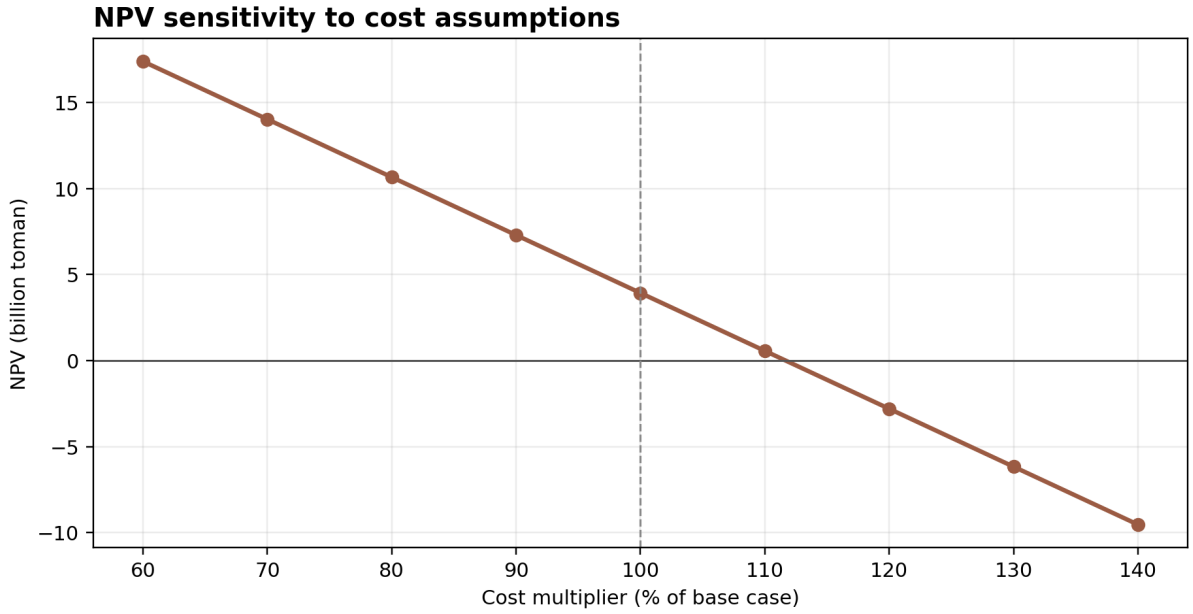


Figure 6: Reported NPV response to cost multipliers. NPV values are in billion toman.

Table 7: Cost sensitivity reported in the PDF

Cost multiplier	NPV (billion toman)
0.60	17.385
0.70	14.020
0.80	10.655
0.90	7.290
1.00	3.925
1.10	0.561
1.20	-2.804
1.30	-6.169
1.40	-9.534

The source report attributes this vulnerability to fixed or semi-fixed obligations such as rent, debt service, maintenance, and other operating expenses. The practical implication is that cost discipline should be a primary management priority. Lease negotiations, purchasing decisions, preventive maintenance, energy efficiency, and staffing productivity can protect the project's economic margin.

17 Combined Scenarios and Risk Interpretation

A two-factor analysis varies revenue and cost multipliers simultaneously. This is closer to real operating conditions than changing one variable at a time, because weak demand may coincide with cost inflation and strong demand may coincide with better utilization. The PDF reports that the real IRR reaches approximately 94% in a favourable case (revenue at 130% and costs at 70%) and falls to about 21% in an adverse case (revenue at 70% and costs at 130%). The adverse result is below the MARR of 37%.

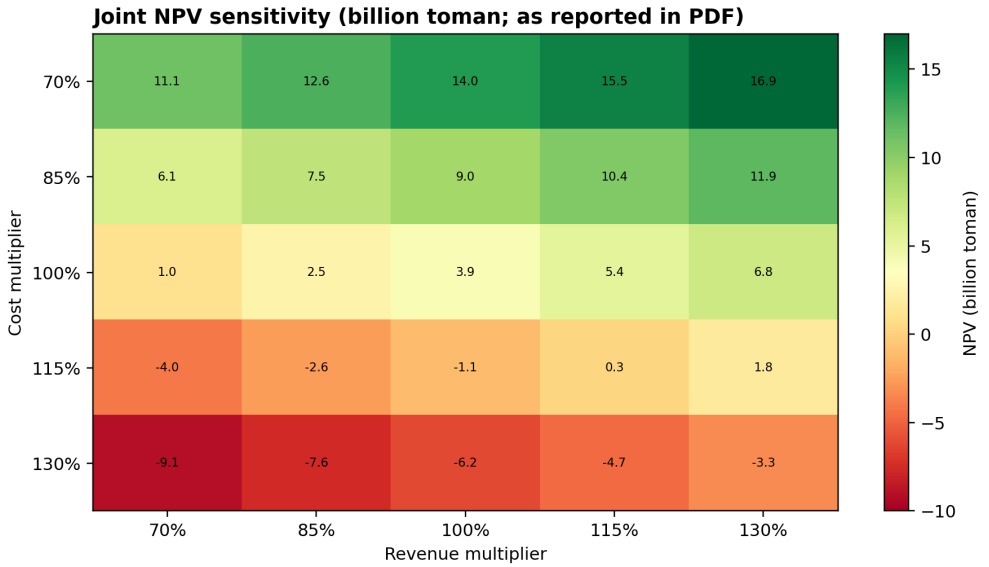


Figure 7: Joint NPV sensitivity matrix from the written report. Rows represent cost multipliers and columns represent revenue multipliers.

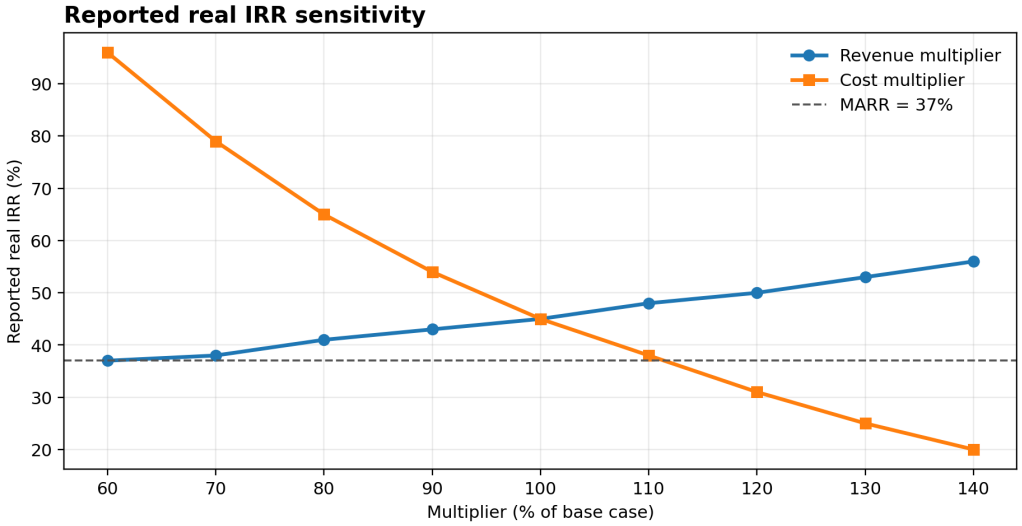


Figure 8: Reported real-IRR sensitivity to revenue and cost multipliers.

The joint analysis emphasizes that the project is not protected by a positive base-case result alone. A simultaneous revenue shortfall and cost overrun can move the model into an economically unacceptable region. In the source report, the main risks are market/revenue risk, cost escalation, inflation, and financing/liquidity risk.

The report recommends ongoing marketing, service quality, cost monitoring, energy management, supplier negotiation, maintenance planning, and adequate cash reserves. These actions are consistent with the sensitivity findings because they target the assumptions with the greatest potential to affect cash flows.

18 Limitations, Reproducibility, and Conclusion

The model provides a structured engineering-economics assessment, but its results depend on estimates rather than observed operating history. Class participation and tuition are assumed; the 40% annual tuition increase compounds over the horizon; utility costs are estimated from assumed usage and tariff conditions; rent is escalated at 40%; and the terminal recovery of the deposit depends on the lease arrangement. These inputs should be stress-tested before using the outputs as an investment recommendation.

A material reproducibility issue was identified when comparing the two supplied files: the workbook's cached sensitivity-table values do not fully match the sensitivity tables and narrative in the PDF, although the base-case NPV and reported IRR can be traced to the main workbook calculations. For this reason, this summary uses the PDF's published sensitivity values and flags the need to recalculate the workbook's what-if tables in Excel. The sensitivity analysis should be treated as provisional until this reconciliation is complete.

Conclusion. Under the source report's base-case assumptions, the project is economically acceptable: NPV is about 3.925 billion toman, reported real IRR is about 45.1%, and payback is about 4.96 years, compared with a MARR of 37%. However, the sensitivity analysis shows that the project is vulnerable to cost increases and to large revenue shortfalls. Its long-term outcome depends on converting projected class demand into actual paying membership while maintaining strict control over rent, operating costs, and debt-service liquidity.

Primary materials used

- *Financial Feasibility Analysis of a Multi-Purpose Sports Club*, original Persian project report, Engineering Economics course project, Sharif University of Technology, Spring 1405.
- *Financial Feasibility Analysis of a Multi-Purpose Sports Club.xlsx*, supplied Excel workbook containing assumptions, operating model, loan amortization, cash-flow projections, economic indicators, and sensitivity tables.